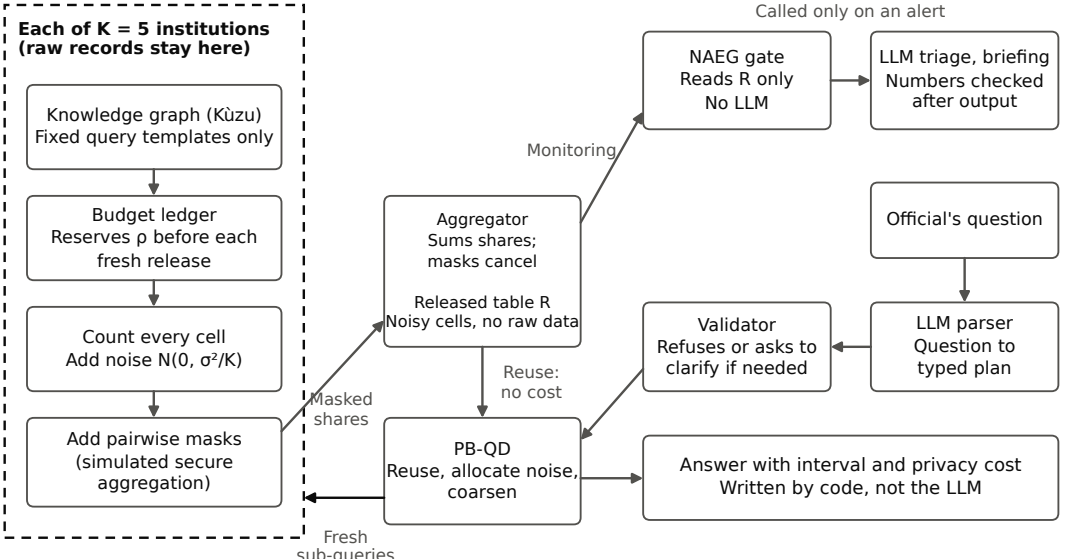
Each of K = 5 institutions
(raw records stay here)
Knowledge graph (Kùzu)
Fixed query templates only
Budget ledger
Reserves ρ before each
fresh release
Count every cell
Add noise N(0, σ²/K)
Add pairwise masks
(simulated secure
aggregation)
Masked
shares
Aggregator
Sums shares;
masks cancel
Released table R
Noisy cells, no raw data
Monitoring
Called only on an alert
NAEG gate
Reads R only
No LLM
LLM triage, briefing
Numbers checked
after output
Official's question
LLM parser
Question to
typed plan
Validator
Refuses or asks to
clarify if needed
Reuse:
no cost
PB-QD
Reuse, allocate noise,
coarsen
Answer with interval and privacy cost
Written by code, not the LLM
Fresh
sub-queries